Q4102 Financial Modelling and Computation

Chapter 1: Lattice Methods

Binomial Tree Method

Option Replication: Replicate option value V with Δ units of underlying + $(V-\Delta S)$ units risk-free, choose Δ such that next period value is deterministic ($V^u = V^d$)

1. **Setup:** Choose portfolio $V = \Delta S + (V - \Delta S)$ in underlying and risk-free
2. **Next Period Value:** $V_{t+\Delta t} = \begin{cases} V^u_{t+\Delta t} = \Delta S_t u + (V-\Delta S)e^{r\Delta t} \\ V^d_{t+\Delta t} = \Delta S_t d + (V-\Delta S)e^{r\Delta t} \end{cases}, \Delta = \frac{V^u_{t+\Delta t} - V^d_{t+\Delta t}}{S_t u - S_t d}$

Delta Hedging: Portfolio of 1 unit option, Δ units of underlying, make it risk-free ($\pi^u = \pi^d$)

1. **Setup:** Choose portfolio $\pi = V - \Delta S$
2. **Next Period Value:** $\pi_{t+\Delta t} = \begin{cases} \pi^u_{t+\Delta t} = V^u_{t+\Delta t} - \Delta S u \\ \pi^d_{t+\Delta t} = V^d_{t+\Delta t} - \Delta S d \end{cases}, \Delta = \frac{V^u_{t+\Delta t} - V^d_{t+\Delta t}}{S_t u - S_t d}$

Solutions: Δ needs to be chosen every period

$$u = e^{\sigma\sqrt{\Delta t}}, d = \frac{1}{u}$$

$$\Delta_t = \frac{V^u_{t+\Delta t} - V^d_{t+\Delta t}}{S_t u - S_t d}, \quad p = \frac{e^{r\Delta t} - d}{u - d}$$

$$V_{t+\Delta t} = e^{-r\Delta t}[pV^u_{t+\Delta t} + (1-p)V^d_{t+\Delta t}] = e^{-r\Delta t}\mathbb{E}^Q_t[V_{t+\Delta t}]$$

Algorithm: Recombining Binomial Tree (Non-Path Dependent Option)

Algorithm 1: *European/American Option Binomial Tree Pricing*

Inputs: $S_0, X, r, T, \sigma, q, N$

Compute: $\Delta t = \frac{T}{N}, u = e^{\sigma\sqrt{\Delta t}}, d = \frac{1}{u}, p = \frac{e^{r\Delta t} - d}{u - d}$

for $j = 0$ to N do:
- $V_N^j = \varphi(S_0 u^{2j-N})$

for $n = N-1$ to 0 do:
- for $j = 0$ to n do:
  - $V_n^j = \max\{e^{-r\Delta t}[pV_n^{j+1} + (1-p)V_n^j], \varphi(S_0 u^{2j-n})\}$

Output: V_0^0

Weakly Path Dependent Options

Barrier Options: Becomes vanilla option depending on whether barrier H is hit

Problem: Slow convergence -> Choose price level i which barrier should align, derive N

$$N = \left\lfloor T\left(\frac{i\sigma}{\ln\left(\frac{H}{S_0}\right)}\right)^2\right\rfloor$$

Algorithm: Recombining Binomial Tree (Set relevant nodes to 0)

Algorithm 2: *European/American Up-and-Out Binomial Tree Pricing*

Inputs: $S_0, X, r, T, \sigma, q, N, H$

Compute: $\Delta t = \frac{T}{N}, u = e^{\sigma\sqrt{\Delta t}}, d = \frac{1}{u}, p = \frac{e^{(r-q)\Delta t} - d}{u - d}$

for $j = 0$ to N do:
- $S_N^j = S_0 u^{2j-N}$
- $V_N^j = \varphi(S_N^j)$ if $S_N^j < H$, else 0

for $n = N-1$ to 0 do:
- for $j = 0$ to n do:
  - $V_n^j = \max\{e^{-r\Delta t}[pV_n^{j+1} + (1-p)V_n^j], \varphi(S_0 u^{2j-n})\}$ if $S_n^j < H$, else 0

Output: V_0^0

Strongly Path Dependent Options

Asian Options: Payoff depends on path average, can be American or European style.

$\varphi(S,A) = \begin{cases} (S-A)^+, fl\ strike\ c \\ (A-S)^+, fl\ strike\ p \\ (A-X)^+, fix\ strike\ c \\ (X-A)^+, fix\ strike\ p \end{cases}$	Arithmetic Average: $A_n = \frac{1}{n+1}\sum_{m=0}^{n} S_{t_m} = \int_0^t S_\tau\, d\tau$ Geometric Average: $A_n = \left(\prod_{m=0}^{n} S_{t_m}\right)^{\frac{1}{n+1}} = \exp\left(\frac{1}{t}\int_0^t \ln S_\tau\, d\tau\right)$

Lookback Option: Payoff depends on path max/min, can be American or European style.

$\varphi(S,A) = \begin{cases} (S-A)^+, fl\ strike\ c \\ (A-S)^+, fl\ strike\ p \\ (A-X)^+, fix\ strike\ c \\ (X-A)^+, fix\ strike\ p \end{cases}$	Running Max: $A_n = \max_{0\le\tau\le n} S_\tau$ Running Min: $A_n = \min_{0\le\tau\le n} S_\tau$

Two-State Variable Binomial Tree Algorithm:

Algorithm 3: Two-State Variable Arith./Geom. Asian or Max/Min Lookback European/American Option

Inputs: $S_0, X, r, T, \sigma, q, N$

Compute: $\Delta t = \frac{T}{N}, u = e^{\sigma\sqrt{\Delta t}}, d = \frac{1}{u}, p = \frac{e^{(r-q)\Delta t} - d}{u - d}, A_0^0 = S_0, S_0^0 = S_0$

for $n = 1$ to N do:
- for $j = 0$ to $2^n - 1$ do:
  - $i = floor\left(\frac{j}{2}\right)$
  - $S_n^j = S_{n-1}^j d$ if j is even
  - $S_n^j = S_{n-1}^j u$ if j is odd
  - $A_n^j = \frac{(n \times A_{n-1}^i + S_n^j)}{n+1} = \exp\left(\frac{1}{n+1}[n \ln A_{n-1}^k + \ln S_{n+1}^j]\right) = \max(A_{n-1}^i, S_n^j) = \min(A_{n-1}^i, S_n^j)$

for $j = 0$ to $2^N - 1$ do:
- $V_N^j = \varphi(S_N^j, A_N^j)$

for $n = N-1$ to 0 do:
- for $j = 0$ to $2^n - 1$ do:
  - $V_n^j = \max\{e^{-r\Delta t}[pV_{n+1}^{j+1} + (1-p)V_{n+1}^j], \varphi(S_0 u^{2j-n})\}$

Output: V_0^0

Similarity Reduction: If $V(S,A,t)$ satisfies $V(\lambda S, \lambda A, t) = \lambda V(S,A,t)$, choose $\lambda = 1/S$

$$\frac{V(S,A,t)}{S} = V\left(1, \frac{A}{S}, t\right) = V(1, e^{x_t}, t) = W(x_t, t)$$

Where $x_t = \ln(A/S)$ then terminal condition reduces to

$$W(x_T, T) = \begin{cases} (e^{x_T} - 1)^+, & fl\ strike\ c \\ (1 - e^{x_T})^+, & fl\ strike\ p \end{cases}$$

and recurrence relation where $\Delta x = \sigma\sqrt{\Delta t} = \ln u, x_n^i = i\Delta x$

$$W(x_n^k, t_n) = e^{-r\Delta t}[puW(x_{n+1}^{2k}, t_{n+1}) + (1-p)dW(x_{n+1}^{2k-1}, t_{n+1})]$$

To convert back, if newly issued just output V_0 if not, interpolate W_0^j

$$V_0 = S_0 W(x_0, t)$$

One-State Variable Binomial Tree Algorithm: *Only works for **FLOATING** strike, reduces number of nodes for Lookback option, does **NOT** reduce number of nodes for Asian.

Algorithm 4: One-State Variable Max/Min Float-Strike Lookback Option *(new/not new)*

Inputs: $S_0, X, r, T, \sigma, q, N, run$

Compute: $\Delta t = \frac{T}{N}, u = e^{\sigma\sqrt{\Delta t}}, d = \frac{1}{u}, p = \frac{e^{(r-q)\Delta t} - d}{u - d}, \Delta x = \ln u, x_0 = \left|\ln\left(\frac{run}{S_0}\right)\right|, j = \frac{x_0}{\Delta x}, j^d = \left\lfloor\frac{x_0}{\Delta x}\right\rfloor$

for $i = 0$ to N do: (for $i = \max(j^d - N, 0)$ to $N + j^d + 1$ do)
- $x_N^i = i\Delta x = -i\Delta x$
- $W_N^i = e^{x_N^i} - 1 = 1 - e^{x_N^i}$

for $n = N-1$ to 0 do:
- for $j = 0$ to n do: (for $i = \max(j^d - n, 0)$ to $n + j^d + 1$ do:)
  - $W_n^j = e^{-r\Delta t}[puW_{n+1}^{(i-1)^+} + (1-p)dW_{n+1}^{i+1}]$
  - $W_n^j = e^{-r\Delta t}[puW_{n+1}^{i+1} + (1-p)dW_{n+1}^{(i-1)^+}]$

Output: $V_0^0 = S_0 W_0^0 \left(= S_0 \times W_0^j \left(W_0^j \coloneqq linearinterpolate\left(j, j^d, j^d+1, W_0^{j^d}, W_0^{j^d+1}\right)\right)\right)$

$$H(\xi, t - \Delta t) = \frac{1}{S}V(S,A,t-\Delta t) = e^{-r\Delta t}[puH(\xi^u, t) + (1-p)dH(\xi^d, t)]$$

Algorithm 6: One-State Variable Geometric Floating Strike Put Asian Option

Inputs: $S_0, X, r, T, \sigma, q, N$

Compute: $\Delta t = \frac{T}{N}, u = e^{\sigma\sqrt{\Delta t}}, d = \frac{1}{u}, p = \frac{e^{(r-q)\Delta t} - d}{u - d}$

for $n = 0$ to N do:
- for $i = 0$ to $2^n - 1$ do:
  - $H_N^i = u^i - 1$

for $n = N-1$ to 0 do:
- for $i = $ N $- 1$ to 0 do:
  - $H_n^i = max(e^{-r\Delta t}[puH_{n+1}^{2k} + (1-p)dH_{n+1}^{2k-1}], (u^i - 1)^+)$

Output: $V_0 = S_0 \times H_0^0$

Forward Shooting Grid (FSG)

Algorithm: Useful when similarity reduction is unavailable (fixed-strike Asian) or ineffective (Asian Option). L is increase in number of grid values per time step.

Algorithm 7: One-State Variable Forward Shooting Grid for Geometric Asian Put

Inputs: $S_0, X, r, T, \sigma, q, N, L$

Compute: $\Delta t = \frac{T}{N}, \rho = \frac{1}{L}, \Delta x = \sigma\sqrt{\Delta t}$

for $k = -LN$ to LN do:
- $x_N^k = k\rho\Delta x$
- $W_N^K = \varphi(x_k^N) = (e^{x_N^k} - 1)^+$

for $n = N-1$ to 0 do:
- for $k = -Ln$ to Ln do:
  - $x_n^k = k\rho\Delta x$
  - Get $x_{n+1}^{k^u}, x_{n+1}^{k^d}$ from x_n^k (*A:$x_{n+1}^{k^{u/d}} = \ln\left(\frac{(n+1)\exp(x_n^k)d/u+1}{n+2}\right)$*G:$\frac{n+1}{n+2}(x_n^k - +\Delta x)$)
  - Compute $k_{n+1}^u = \frac{x^{k^u}}{\rho\Delta x}, k_{n+1}^d = \frac{x^{k^d}}{\rho\Delta x}, k_{cl}^u, k_{fl}^u, k_{cl}^d, k_{fl}^d$
  - Compute $W_{n+1}^{k^u}$ and $W_{n+1}^{k^d}$ with linear interpolation
  - $W_n^k = e^{-r\Delta t}[puW_{n+1}^{k^u} + (1-p)dW_{n+1}^{k^d}]$

Output: $V_0^0 = S_0 \times W_0^j \left(W_0^j \coloneqq linearinterpolate\left(j, j^d, j^d+1, W_0^{j^d}, W_0^{j^d+1}\right)\right)$

Algorithm 8: Two-state-variable FSG Method for fixed-strike arithmetic Asian call option (newly-issued/not new)

Inputs: $S_0, X, r, T, \sigma, q, N, L, A_{run}, N_{hist}$

Compute: $\Delta t = \frac{T}{N}, u = e^{\sigma\sqrt{\Delta t}}, p = \frac{e^{(r-q)\Delta t} - d}{u - d}, \Delta x = \ln u, \rho = \frac{1}{L}, A_0 = \frac{N_{hist}A_{run} + S_0}{N_{hist} + 1}$

for $n = 0$ to N do:
- for $k = -NL$ to NL do:
  - $A_N^{(i,k)} = S_0 e^{k\rho\Delta x} = A_0 e^{k\rho\Delta x}$
  - $V_N^{(i,k)} = (A_N^{(i,k)} - X)^+$

for $n = N-1$ to 0 do:
- for $i = $ N $- 1$ to 0 do:
  - $S_i^t = S_0 u^i d^{n-i}$
  - for $k = -tL$ to tL do
    - $A_n^{(i,k)} = S_0 e^{k\rho\Delta x}, A_n^{(i,k)} = A_0 e^{k\rho\Delta x}$
    - $A^u = \frac{(S_t^i u) + (N_{hist} + n + 1)A_t^{(i,k)}}{N_{hist} + t + 2}, A^d = \frac{(S_t^i d) + (N_{hist} + n + 1)A_t^{(i,k)}}{N_{hist} + t + 2}$
    - $k_{cl}^u = \left\lceil\frac{\ln(A_u/S_0)}{\rho\Delta x}\right\rceil, k_{fl}^u = \left\lfloor\frac{\ln(A_u/S_0)}{\rho\Delta x}\right\rfloor, k_{cl}^d = \left\lceil\frac{\ln(A_d/S_0)}{\rho\Delta x}\right\rceil, k_{fl}^d = \left\lfloor\frac{\ln(A_d/S_0)}{\rho\Delta x}\right\rfloor$
    - $k_{cl}^u = \left\lceil\frac{\ln(A_u/A_0)}{\rho\Delta x}\right\rceil, k_{fl}^u = \left\lfloor\frac{\ln(A_u/A_0)}{\rho\Delta x}\right\rfloor, k_{cl}^d = \left\lceil\frac{\ln(A_d/A_0)}{\rho\Delta x}\right\rceil, k_{fl}^d = \left\lfloor\frac{\ln(A_d/A_0)}{\rho\Delta x}\right\rfloor$
    - *Compute* V^u, V^d *using LinearInterpolation*
    - $V_n^{(i,k)} = e^{-r\Delta t}(pV^u + (1-p)V^d)$

Output: $V_0^{(0,0)}$

Chapter 2: Continuous Time Models

Wiener Process

Definition A stochastic process $W = (W_t)_{t\ge0}$ is called a Wiener Process/Brownian Motion if (1) it starts at 0, (2) with probability 1 the function $t \to W_t$ is continuous, (3) it has stationary and (4) independent increments, and (5) increments $W_{t+s} - W_t \sim N(0,s)$

Variation $V_t^P(W) = \sum_{i=1}^n |W_{t_i} - W_{t_{i-1}}|$, $V_t^P(W) \cong n\sqrt{t/n}$, as $n \to \infty, V_t(W) \to \infty$

Quadratic Variation $Q_t^P(W) = \sum_{i=1}^n |W_{t_i} - W_{t_{i-1}}|^2$, $Q_t^P = \sum \Delta W_{t_i}^2 \to t$, $\int_0^t (dW_s)^2 = \int_0^t ds = t$

$$dW_t dW_t = dt, \quad dtdt = 0, \quad dW_t dt = 0$$

Ito's Lemma: $dV_t = \frac{\partial V}{\partial t}dt + \frac{\partial V}{\partial S}dS_t + \frac{1}{2}\frac{\partial^2 V}{\partial S^2}dS_t^2$

Multivariable Ito: $dV = \frac{\partial V}{\partial t}dt + \sum_{i=0}^n \frac{\partial V}{\partial S^i}dS^i + \frac{1}{2}\sum_{i,j=0}^n \frac{\partial^2 V}{\partial S^i \partial S^j}dS^i dS^j$

Black-Scholes Framework

1. $dS_t = \mu S\, dt + \sigma S\, dW_t$
2. r, σ assumed constant over life span of option
3. Trading is continuous, no short selling, fractional qty of assets ok
4. No transaction cost, no taxes
5. No arbitrage opportunities

$$\mathcal{L}_{BS} = \frac{\partial}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2}{\partial S^2} + (r-q)S\frac{\partial}{\partial S} - r = 0$$

PDE for European Options

Option on S, price $V = V(S,t)$, $S \to qS\,dt$ in dividends

Port: long option, hedge position w Δ shares, $\Pi = V - \Delta S$

$$d\Pi_t = dV_t - \Delta dS_t - q\Delta S\,dt = \left(\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - q\Delta S\right)dt + \frac{\partial V}{\partial S}dS_t - \Delta\, dS_t$$

Make port riskless, choose $\Delta = \frac{\partial V}{\partial S}$. By no-arbit, earns risk-free rate

$$d\Pi_t = r\Pi\,dt = r(V - \Delta S)\,dt = r\left(V - \frac{\partial V}{\partial S}S\right)dt = \left(\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2\frac{\partial^2 V}{\partial S^2} - q\frac{\partial V}{\partial S}S\right)dt$$

$$\therefore \frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + (r-q)S\frac{\partial V}{\partial S} - rV = \mathcal{L}_{BS}V = 0$$

Solution domain: $D = \{(S,t): S > 0, 0 \le t < T\}, V(S,T) = \begin{cases}(S-X)^+ \\ (X-S)^+\end{cases}$

$$V(S,t) = \begin{cases} Se^{-q(T-t)}N(d_1) - Xe^{-r(T-t)}N(d_2) & call \\ Xe^{-r(T-t)}N(-d_2) - Se^{-q(T-t)}N(-d_1) & put \end{cases}, \quad d_1 = \frac{\ln\frac{S}{X} + \left(r - q + \frac{\sigma^2}{2}\right)(T-t)}{\sigma\sqrt{T-t}}$$

$$d_2 = d_1 - \sigma\sqrt{T-t}$$

PDE for American Options

- Call with no dividends → never optimal to exercise early
- Exist optimal exercise boundary $S^*(t)$, separate continuation region (H) and early-exercise region (E)

In H: $\begin{matrix} S > S^*, put \\ S < S^*, call \end{matrix}, \mathcal{L}_{BS}V = 0$, In E: $\begin{matrix} S \le S^*, put \\ S > S^*, call \end{matrix}$ $V = \varphi(S) = \begin{matrix} X - S \\ S - X \end{matrix}$

Either region: $V(S,t) \ge \varphi(S)$, **Δ hedged:** $d\Pi_t = \left(\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2\frac{\partial^2 V}{\partial S^2} - q\frac{\partial V}{\partial S}S\right)dt$

Long Short Asymmetry: $d\Pi_t \le r\Pi_t\,dt$, $\mathcal{L}_{BS}V \le 0$

Linear Complementarity:

$$\begin{cases} \mathcal{L}_{BS}V \le 0 \\ V - \varphi(S) \ge 0 \\ [\mathcal{L}_{BS}V][V - \varphi(S)] = 0 \end{cases}, \quad \begin{matrix} D = \{(S,t): S > 0, 0 \le t < T\} \\ V(S,T) = \varphi(S)^+ \end{matrix}$$

PDE for Asian Options

$$A: dA_t = \frac{1}{t}(S - A)\,dt, \quad G: dA_t = \frac{A}{t}\ln\frac{S}{A}\,dt$$

$$dV(S,A,t) = \frac{\partial V}{\partial t}dt + \frac{\partial V}{\partial S}dS + \frac{1}{2}\frac{\partial^2 V}{\partial S^2}(dS)^2 + \frac{\partial V}{\partial A}dA$$

$$d\Pi_t = dV - \Delta dS_t - q\Delta S dt = \left(\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2\frac{\partial^2 V}{\partial S^2} - q\Delta S\right)dt + \frac{\partial V}{\partial A}dA_t + \frac{\partial V}{\partial S}dS_t - \Delta\, dS_t$$

Make port riskless, choose $\Delta = \frac{\partial V}{\partial S}$. By no-arbit, earns risk-free rate

$$d\Pi_t = r\Pi\,dt = r\left(V - \frac{\partial V}{\partial S}S\right)dt = \left(\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2\frac{\partial^2 V}{\partial S^2} - q\frac{\partial V}{\partial S}S\right)dt + \frac{\partial V}{\partial A}dA_t$$

$$\therefore \frac{\partial V}{\partial t} + g(S,A,t)\frac{\partial V}{\partial A} + \frac{1}{2}\sigma^2 S^2\frac{\partial^2 V}{\partial S^2} + (r-q)S\frac{\partial V}{\partial S} - rV = \mathcal{L}_{BS}V + g(S,A,t)\frac{\partial V}{\partial A} = 0$$

Where $g(S,A,t) = \begin{cases} \frac{A}{t}\ln\frac{S}{A}, cont\ geo \\ \frac{1}{t}(S-A), cont\ arith \end{cases}$

$$\begin{cases} \mathcal{L}_a V = 0, & S > 0, A > 0, t \in [0,T) \\ V(S,A,T) = \varphi(S,A) \end{cases}, \quad where\ \mathcal{L}_a V \begin{cases} \frac{1}{t}(S-A)\frac{\partial V}{\partial A} + \mathcal{L}_{BS}V \\ \frac{A}{t}\ln\frac{S}{A}\frac{\partial V}{\partial A} + \mathcal{L}_{BS}V \end{cases}$$

PDE for Lookback Options

Append path state variable M or m

Infinity norm: $\|\vec{y}\|_\infty = \lim_{n\to\infty}\|\vec{y}\|_n = \lim_{n\to\infty}(\sum_{k=1}^p |y_k|^n)^{\frac{1}{n}} = \max\{y_k\}$ limit

Maximum as Limit: $M_n = \left[\int_0^t (S_u)^n\, du\right]^{\frac{1}{n}}$, $dM_n = \frac{1}{n}\frac{S^n}{(M_n)^{n-1}}dt$

$$dV = \frac{\partial V}{\partial t}dt + \frac{\partial V}{\partial S}dS + \frac{1}{2}\frac{\partial^2 V}{\partial S^2}(dS)^2 + \frac{\partial V}{\partial M_n}dM_n$$

Minimum as Limit: $m_n = \left[\int_0^t S_u^{-n}\, du\right]^{-\frac{1}{n}}$, $dm_n = -\frac{1}{n}\frac{1}{(m_n)^{1-n}S^n}dt$

$$dV = \frac{\partial V}{\partial t}dt + \frac{\partial V}{\partial S}dS + \frac{1}{2}\frac{\partial^2 V}{\partial S^2}(dS)^2 + \frac{\partial V}{\partial m_n}dm_n$$

Use delta-hedging strat, choose $\Delta = \frac{\partial V}{\partial S}$ (for max as limit)

$$\frac{\partial V}{\partial t} + \frac{1}{n}\frac{S^n}{(M_n)^{n-1}}\frac{\partial V}{\partial M_n} + \frac{1}{2}\sigma^2 S^2\frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0$$

$$PDE\ fl\ lookback\ max = \begin{cases} \mathcal{L}_{BS}V(S,M,t) = 0, 0 < S < M, 0 < t < T \\ V(0,M,t) = e^{-r(T-t)}M \\ \frac{\partial V(M,M,t)}{\partial M} = 0 \\ V(S,M,T) = M - S \end{cases}$$

Consistency of BTM and BS PDE *(*NOTE: Consistency ≠ Convergence)*

Taylor Series: $\sum_{n=0}^\infty \frac{f^{(n)}(a)}{n!}(x-a)^n$

Vanilla options:

BTM: $V(S, t - \Delta t) = e^{-r\Delta t}[pV(Su,t) + (1-p)V(Sd,t)]$

BS PDE: $\mathcal{L}_{BS}V = \frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2\frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0$

$$V(Su,t) = V(S,t) + (u-1)S\frac{\partial V}{\partial S} + \frac{(u-1)^2}{2}S^2\frac{\partial^2 V}{\partial S^2} + \frac{(u-1)^3}{6}S^3\frac{\partial^3 V}{\partial S^3} + O[(\Delta t)^2]$$

$$V(Sd,t) = V(S,t) + (d-1)S\frac{\partial V}{\partial S} + \frac{(d-1)^2}{2}S^2\frac{\partial^2 V}{\partial S^2} + \frac{(d-1)^3}{6}S^3\frac{\partial^3 V}{\partial S^3} + O[(\Delta t)^2]$$

$$V(S, t-\Delta t) = V(S,t) + \Delta t\frac{\partial V}{\partial t} + \frac{(\Delta t)^2}{2}\frac{\partial^2 V}{\partial t^2} + O[(\Delta t)^3]$$

$$\begin{cases} e^{-r\Delta t}[p(u-1) + (1-p)(d-1)] = r\Delta t + O[(\Delta t)^2], & BTM = risk\ free \\ e^{-r\Delta t}[p(u-1)^2 + (1-p)(d-1)^2] = \sigma^2\Delta t + O[(\Delta t)^2], & variance\ relation \\ e^{-r\Delta t}[p(u-1)^3 + (1-p)(d-1)^3] = O[(\Delta t)^2], & \frac{d-1}{d} = 1 - u \end{cases}$$

Apply the formulas above: LHS reduces to RHS

$$\frac{-V(S,t-\Delta t) + e^{-r\Delta t}[pV(Su,t) + (1-p)V(Sd,t)]}{\Delta t} = \mathcal{L}_{BS}V + O(\Delta t)$$

American vanilla options:

BTM: $V(S,t-\Delta t) = \max(e^{-r\Delta t}[pV(Su,t) + (1-p)V(Sd,t)], \varphi(S))$

BS PDE: $\begin{cases} \mathcal{L}_{BS}V \le 0 \\ V - \varphi(S) \ge 0 \\ [\mathcal{L}_{BS}V][V - \varphi(S)] = 0 \end{cases} \to \min\{-\mathcal{L}_{BS}V, V - \varphi(S)\} = 0$

Similarly:

$$0 = \min(V(S,t-\Delta t) - e^{-r\Delta t}[pV(Su,t) + (1-p)V(Sd,t)], V(S,t-\Delta t) - \varphi(S))$$
$$= \min(-\mathcal{L}_{BS}V\Delta t + O(\Delta t^2), V - \varphi(S) + O(\Delta t))$$
$$= \min\{-\mathcal{L}_{BS}V + O(\Delta t), V - \varphi(S) + O(\Delta t)\} = 0$$

Barrier Options:

$$\begin{cases} \mathcal{L}_{BS}V = 0, & (S,t) \in D \\ V(H,t) = 0, & 0 \le t < T \\ V(S,T) = \varphi(S)^+, & 0 < S < H \end{cases}, \begin{cases} V(S,t-\Delta t) = e^{-r\Delta t}[pV(Su,t) + (1-p)V(Sd,t)], \\ V(S,t) = 0, \quad 0 \le t < T \\ V(S,T) = \varphi(S)^+, \quad 0 < S < H \end{cases}$$

Asian Options:

BTM: $V(S,A,t-\Delta t) = e^{-r\Delta t}[pV(Su,A^u,t) + (1-p)V(Sd,A^d,t)]$

BS PDE: $\begin{cases} \mathcal{L}_a V = 0, \ S > 0, A > 0, t \in [0,T) \\ V(S,A,T) = \varphi(S,A) \end{cases}$, where $\mathcal{L}_a V \begin{cases} \frac{1}{t}(S-A)\frac{\partial V}{\partial A} + \mathcal{L}_{BS}V \\ \frac{A}{t}\ln\frac{S}{A}\frac{\partial V}{\partial A} + \mathcal{L}_{BS}V \end{cases}$

$$V(Su,A^u,t) = V(S,A,t) + (u-1)S\frac{\partial V}{\partial S} + \frac{(u-1)^2}{2}S^2\frac{\partial^2 V}{\partial S^2} + \frac{(u-1)^3}{6}S^3\frac{\partial^3 V}{\partial S^3} + \cdots$$
$$+(A^u - A)\frac{\partial V}{\partial A} + \frac{(A^u-A)^2}{2!}\frac{\partial^2 V}{\partial A^2} + \frac{2(u-1)(A^u-A)}{2!}S\frac{\partial^2 V}{\partial A\partial S} + \cdots$$

$$V(Sd,A^d,t) = V(S,A,t) + (d-1)S\frac{\partial V}{\partial S} + \frac{(d-1)^2}{2}S^2\frac{\partial^2 V}{\partial S^2} + \frac{(d-1)^3}{6}S^3\frac{\partial^3 V}{\partial S^3} + \cdots$$
$$+(A^d - A)\frac{\partial V}{\partial A} + \frac{(A^d-A)^2}{2!}\frac{\partial^2 V}{\partial A^2} + \frac{2(d-1)(A^d-A)}{2!}S\frac{\partial^2 V}{\partial A\partial S} + \cdots$$

$$V(S,A,t-\Delta t) = V(S,A,t-\Delta t) - \Delta t\frac{\partial V}{\partial t} + \frac{(\Delta t)^2}{2}\frac{\partial^2 V}{\partial t^2} + O[(\Delta t)^3]$$

$$\begin{cases} e^{-r\Delta t}[p(A^u - A) + (1-p)(A^d - A)] = \frac{S-A}{t}\Delta t + O[(\Delta t)^2], & How? \\ e^{-r\Delta t}[p(A^u-A)^2 + (1-p)(A^d-A)^2] = O[(\Delta t)^2], & How? \\ e^{-r\Delta t}[p(u-1)(A^u-A) + (1-p)(d-1)(A^d-A)] = O[(\Delta t)^2], & How? \end{cases}$$

Apply the formulas above: LHS reduces to RHS

$$\frac{-V(S,A,t-\Delta t) + e^{-r\Delta t}[pV(Su,A^u,t) + (1-p)V(Sd,A^d,t)]}{\Delta t} = \mathcal{L}_a V + O(\Delta t)$$

Lookback Options:

BTM: $V(S,A,t-\Delta t) = e^{-r\Delta t}[pV(Su,\max(A,Su)) + (1-p)V(Sd,A,t)]$

BS PDE: $\begin{cases} \mathcal{L}_{BS}V(S,M,t) = 0, \ 0 < S < M, 0 < t < T \\ V(0,M,t) = e^{-r(T-t)}M \\ \frac{\partial V(M,M,t)}{\partial M} = 0 \\ V(S,M,T) = M - S \end{cases}$

Case 1: $S < A, Su < A$

The Binomial pricing equation becomes

$$V(S,A,t-\Delta t) = e^{-r\Delta t}[pV(Su,A,t) + (1-p)V(Sd,A,t)]$$

which is consistent with the PDE, $\mathcal{L}_{BS}V = 0$

Case 2: $S = A$

$$V(S,A,t-\Delta t) = e^{-r\Delta t}[pV(Su,Su,t) + (1-p)V(Sd,A,t)]$$

which is consistent with the Neumann boundary condition

Use Taylor series from Asian to prove

$$0 = -V(S,S,t-\Delta t) + e^{-r\Delta t}[pV(Su,Su,t) + (1-p)V(Sd,S,t)]$$
$$= \mathcal{L}_{BS}V\Delta t + O((\Delta t)^2) + \frac{\partial V}{\partial A}e^{-r\Delta t}[pS(u-1)] + O(\Delta t)$$

Useful: $u - 1 = \sigma\sqrt{\Delta t} + \frac{1}{2}\sigma^2\Delta t + \frac{1}{6}\sigma^3\Delta t^{\frac{3}{2}} + O((\Delta t)^2) = \sigma\sqrt{\Delta t} + O(\Delta t)$

$$p = \frac{1}{2} + O(\sqrt{\Delta t})$$

$$\therefore 0 = \mathcal{L}_{BS}V\Delta t + O((\Delta t)^2) + \frac{1}{2}\frac{\partial V}{\partial A}e^{-r\Delta t}S\sigma\sqrt{\Delta t} + O(\Delta t)$$
$$= \mathcal{L}_{BS}V\sqrt{\Delta t} + \frac{1}{2}\frac{\partial V}{\partial A}e^{-r\Delta t}S\sigma + O(\sqrt{\Delta t})$$

As $\Delta t \to 0, \frac{\partial V}{\partial A} \to 0$

Convergence (NOTE: consistency ≠ convergence)A

Convergence of BTM for European options:

$V_n^j = e^{-r\Delta t}[pV_{n+1}^{j+1} + (1-p)V_{n+1}^j]$ is BTM solution

$v_n^j = e^{-r\Delta t}[pv_{n+1}^{j+1} + (1-p)v_{n+1}^j] + O((\Delta t)^2)$ is cont time solution

$z_n^j = v_n^j - V_n^j = e^{-r\Delta t}[pz_{n+1}^{j+1} + (1-p)z_{n+1}^j] + O[(\Delta t)^2]$ error at n

$$\mathbf{z}_n = e^{-r\Delta t}\mathbf{A}\mathbf{z}_{n+1} + O[(\Delta t)^2]\mathbf{1}$$

Where $\mathbf{A}_{(n+1)\times(n+2)} = \begin{bmatrix} 1-p & p & 0 & \cdots & \cdots & 0 \\ 0 & 1-p & p & 0 & \cdots & 0 \\ 0 & \ddots & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & \cdots & 0 & 1-p & p \end{bmatrix}$

$$\|\mathbf{z}_n\| = \|e^{-r\Delta t}\mathbf{A}\mathbf{z}_{n+1} + O[(\Delta t)^2]\mathbf{1}\| \le e^{-r\Delta t}\|\mathbf{A}\|\|\mathbf{z}_{n+1}\| + O((\Delta t)^2)\|\mathbf{1}\|$$

Note $r > 0, p \in (0,1)$, where $\|\mathbf{z}_n\| = \max_j |z_n^j|$

$$\|\mathbf{z}_n\| \le \|\mathbf{z}_{n+1}\| + C(\Delta t)^2 \le \|\mathbf{z}_{n+2}\| + 2C(\Delta t)^2 \le \cdots$$
$$\le \|\mathbf{z}_N\| + (N-n)C(\Delta t)^2 \le TC\Delta t \to 0\ as\ \Delta t \to 0$$

Convergence of BTM for American options:

BTM for American options is nonlinear, consider scheme $\boldsymbol{Y} = \boldsymbol{GX}$

If scheme has (i): consistency: (ii) monotonicity: if $\boldsymbol{V} \le \boldsymbol{U}, \boldsymbol{GV} \le \boldsymbol{GU}$, (iii) local stability, and comparisons principle holds for PDE models

Convergence of FSGM:

Chapter 3: Finite Differences Method

Discretizing Problem Domain

Discretization: $[0, S_{max}], [0,T]$ into I and N sub-intervals of h and Δt

$$\{(S_n^i, t_n): S_n^i = ih, t_n = n\Delta t, \quad i = 0,1,\ldots,I, \quad n = 0,1,\ldots,N\}$$

Exact option value: $v_n^i = v(S_n^i, t_n)$, **approx option value:** $V_n^i = V(S_n^i, t_n)$

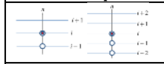
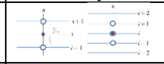
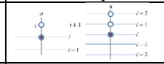
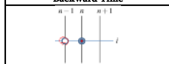
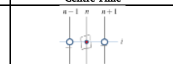
Finite Differences Formula

$$v_n^{i+1} = v_n^i + h\frac{\partial v}{\partial S}|_{S_n^i,t_n} + \frac{1}{2}h^2\frac{\partial^2 h}{\partial S^2}|_{S_n^i,t_n} + \frac{1}{6}h^3\frac{\partial^3 v}{\partial S^3}|_{S_n^i,t_n} + O(h^4)$$

$$v_n^{i-1} = v_n^i - h\frac{\partial v}{\partial S}|_{S_n^i,t_n} + \frac{1}{2}h^2\frac{\partial^2 h}{\partial S^2}|_{S_n^i,t_n} - \frac{1}{6}h^3\frac{\partial^3 v}{\partial S^3}|_{S_n^i,t_n} + O(h^4)$$

$$v_{n+1}^i = v_n^i + \Delta t\frac{\partial v}{\partial t}|_{S_n^i,t_n} + O((\Delta t)^2)$$

$$v_{n-1}^i = v_n^i - \Delta t\frac{\partial v}{\partial t}|_{S_n^i,t_n} + O((\Delta t)^2)$$

Backward Space	Centre Space	Forward Space
		
$\frac{\partial v}{\partial S}\|_{S_n^i,t_n} = \frac{v_n^i - v_n^{i-1}}{h} + O(h)$ $\frac{\partial^2 v}{\partial S^2}\|_{S_n^i,t_n} = \frac{v_n^i - 2v_n^{i+1} + v_n^{i+2}}{h^2} + O(h)$	$\frac{\partial v}{\partial S}\|_{S_n^i,t_n} = \frac{v_n^{i+1} - v_n^{i-1}}{2h} + O(h^2)$ $\frac{\partial^2 v}{\partial S^2}\|_{S_n^i,t_n} = \frac{v_n^{i-1} - 2v_n^i + v_n^{i+1}}{h^2} + O(h^2)$	$\frac{\partial v}{\partial S}\|_{S_n^i,t_n} = \frac{v_n^{i+1} - v_n^i}{h} + O(h)$ $\frac{\partial^2 v}{\partial S^2}\|_{S_n^i,t_n} = \frac{v_n^i - 2v_n^{i+1} + v_n^{i+2}}{h^2} + O(h)$
Backward Time	**Centre Time**	**Forward Time**
		
$\frac{\partial v}{\partial t}\|_{S_n^i,t_n} = \frac{v_n^i - v_{n-1}^i}{\Delta t} + O(\Delta t)$	$\frac{\partial v}{\partial t}\|_{S_n^i,t_n} = \frac{v_{n+1}^i - v_{n-1}^i}{2\Delta t} + O(\Delta t^2)$	$\frac{\partial v}{\partial t}\|_{S_n^i,t_n} = \frac{v_{n+1}^i - v_n^i}{\Delta t} + O(\Delta t)$

Explicit Scheme

Boundary conditions:

(+) no need solve system of equations, ease of implementation

(-) conditionally convergent

$$\frac{\partial v}{\partial t}|_{S_{n+1}^i,t_{n+1}} + \frac{1}{2}\sigma^2 S^2\frac{\partial^2 v}{\partial S^2}|_{S_{n+1}^i,t_{n+1}} + rS\frac{\partial v}{\partial S}|_{S_{n+1}^i,t_{n+1}} - rv|_{S_{n+1}^i,t_{n+1}} = 0$$

Explicit scheme (I):

$$\frac{v_{n+1}^i - v_n^i}{\Delta t} + \frac{1}{2}\sigma^2(S_{n+1}^i)^2\frac{v_{n+1}^{i+1} - 2v_{n+1}^i + v_{n+1}^{i-1}}{h^2} + rS_{n+1}^i\frac{v_{n+1}^{i+1} - v_{n+1}^{i-1}}{2h} - rv_{n+1}^i + O(\Delta t + h^2) = 0$$

$$V_n^i = \frac{1}{2}(\sigma^2 i^2 - ri)\Delta t V_{n+1}^{i-1} + (1 - \sigma^2 i^2\Delta t - r\Delta t)V_{n+1}^i + \frac{1}{2}(\sigma^2 i^2 + ri)\Delta t V_{n+1}^{i+1}, \quad i = 1,2,\ldots,I-1$$

Monotonicity condition: $1 - \sigma^2 i^2\Delta t - r\Delta t > 0$, $\frac{1}{2}(\sigma^2 i^2 - ri) > 0$

Explicit scheme (II):

$$\frac{v_{n+1}^i - v_n^i}{\Delta t} + \frac{1}{2}\sigma^2(S_{n+1}^i)^2\frac{v_{n+1}^{i+1} - 2v_{n+1}^i + v_{n+1}^{i-1}}{h^2} + rS_{n+1}^i\frac{v_{n+1}^{i+1} - v_{n+1}^{i-1}}{2h} - rv_n^i + O(\Delta t) + O(\Delta t + h^2) = 0$$

$$V_n^i = \frac{1}{1 + r\Delta t}\left[\frac{1}{2}(\sigma^2 i^2 - ri)\Delta t V_{n+1}^{i-1} + (1 - \sigma^2 i^2\Delta t)V_{n+1}^i + \frac{1}{2}(\sigma^2 i^2 + ri)\Delta t V_{n+1}^{i+1}\right]$$

Monotonicity condition: $1 - \sigma^2 i^2\Delta t > 0$

Explicit scheme (III):

$$\frac{v_{n+1}^i - v_n^i}{\Delta t} + \frac{1}{2}\sigma^2(S_{n+1}^i)^2\frac{v_{n+1}^{i+1} - 2v_{n+1}^i + v_{n+1}^{i-1}}{h^2} + rS_{n+1}^i\frac{v_{n+1}^{i+1} - v_{n+1}^i}{h} - rv_n^i + O(\Delta t) + O(\Delta t + h) = 0$$

$$V_n^i = \frac{1}{1 + r\Delta t}\left[\frac{1}{2}\sigma^2 i^2\Delta t V_{n+1}^{i-1} + (1 - \sigma^2 i^2\Delta t - ri\Delta t)V_{n+1}^i + \left(\frac{1}{2}\sigma^2 i^2 + ri\right)\Delta t V_{n+1}^{i+1}\right]$$

Monotonicity condition: $1 - \sigma^2 i^2\Delta t - ri\Delta t > 0$

Discretization of terminal and boundary conditions: (put)

Terminal condition: $V_N^i = (X - ih)^+\ for\ i = 0,1,\ldots,I$

At $S = 0$: $V_n^0 = Xe^{-r(T - n\Delta t)}\ for\ n = 0,1,\ldots,N$

At $S = S_{max}$: $V_n^I = 0\ for\ n = 0,1,\ldots,N$

In general: $\boldsymbol{V_n^i = a_i V_{n+1}^{i-1} + b_i V_{n+1}^i + c_i V_{n+1}^{i+1},\ i \in Z_I = 1,2,\ldots,I-1}$

$$\begin{bmatrix} V_n^1 \\ \vdots \\ V_n^{I-1} \end{bmatrix} = \begin{bmatrix} a_1 \\ \vdots \\ a_{I-1} \end{bmatrix} \circ \begin{bmatrix} V_{n+1}^0 \\ \vdots \\ V_{n+1}^{I-2} \end{bmatrix} + \begin{bmatrix} b_1 \\ \vdots \\ b_{I-1} \end{bmatrix} \circ \begin{bmatrix} V_{n+1}^1 \\ \vdots \\ V_{n+1}^{I-1} \end{bmatrix} + \begin{bmatrix} c_1 \\ \vdots \\ c_{I-1} \end{bmatrix} \circ \begin{bmatrix} V_{n+1}^2 \\ \vdots \\ V_{n+1}^I \end{bmatrix}$$

$$\boldsymbol{V_n = AV_{n+1} + F_{n+1}}$$

$$\begin{bmatrix} V_n^1 \\ V_n^2 \\ \vdots \\ V_n^{I-2} \\ V_n^{I-1} \end{bmatrix} = \left[\begin{pmatrix} b_1 & c_1 & & & \\ a_2 & b_2 & c_2 & & \\ & \ddots & \ddots & & \\ & & a_{I-2} & b_{I-2} & c_{I-2} \\ & & & a_{I-1} & b_{I-1} \end{pmatrix}\right]\begin{bmatrix} V_{n+1}^1 \\ V_{n+1}^2 \\ \vdots \\ V_{n+1}^{I-2} \\ V_{n+1}^{I-1} \end{bmatrix} + \begin{bmatrix} a_1 V_{n+1}^0 \\ 0 \\ \vdots \\ 0 \\ c_{I-1}V_{n+1}^I \end{bmatrix}$$

Monotonicity explicit schemes: $\boldsymbol{AU_{n+1} \le AV_{n+1} \to A = (a_{ij}), a_{ij} \ge 0}$

Transformed PDE Explicit Scheme

Explicit scheme for transformed PDE $\quad u(x,t) = v(S,t), x = \ln S$

put: $\frac{\partial u}{\partial t} = \frac{1}{2}\sigma^2\frac{\partial^2 u}{\partial x^2} + \left(r - \frac{\sigma^2}{2}\right)\frac{\partial u}{\partial x} - ru = 0,\ x \in (-\infty,\infty), t \in [0,T), u(x,T) = (X - e^x)^+$

Discretization of domain:

Truncate domain $x \in (-\infty,\infty)$ to $x \in [x_{min}, x_{max}]$ where $e^{x_{min}}, e^{x_{max}}$ sufficiently small/large

$$\{(\boldsymbol{x_n^i, t_n}): \boldsymbol{x_n^i = x_{min} + i\Delta x, t_n = n\Delta t}, \quad \boldsymbol{i = 0,1,\ldots I}, \quad \boldsymbol{n = 0,1,\ldots,N}\}$$

Discretization of PDE: evaluated at (x_{n+1}^i, t_{n+1})

$$\frac{u_{n+1}^i - u_n^i}{\Delta t} + \frac{\sigma^2}{2}\frac{u_{n+1}^{i+1} - 2u_{n+1}^i + u_{n+1}^{i-1}}{(\Delta x)^2} + \left(r - \frac{\sigma^2}{2}\right)\frac{u_{n+1}^{i+1} - u_{n+1}^{i-1}}{2\Delta x} - ru_n^i + O(\Delta t + (\Delta x)^2) = 0$$

$$U_n^i = \frac{1}{1 + r\Delta t}[\alpha(1-a)U_{n+1}^{i-1} + (1-\alpha)U_{n+1}^i + \alpha a U_{n+1}^{i+1}], \quad \alpha = \frac{\sigma^2\Delta t}{(\Delta x)^2}, a = \frac{1}{2} + \left(r - \frac{\sigma^2}{2}\right)\frac{\Delta x}{2\sigma^2}$$

Discretization of terminal and boundary conditions:

$$U_N^i = \left(X - e^{x_N^i}\right)^+, \quad i = 0,1,\ldots,I$$

Monotonicity: $\Delta x\ small, 0 < a < 1, \to (1-\alpha) > 0 \to \Delta t \le \frac{(\Delta x)^2}{\sigma^2}$

Implicit Schemes

(+) good stability/convergence properties

(-) need solve system of equations

Discretization of domain:

Truncate domain $S > 0$ to $S \in [0, S_{max}]$ for S_{max} sufficiently large

$$\{(S_n^i, t_n): S_n^i = ih, t_n = n\Delta t, \quad i = 0,1,\ldots I, \quad n = 0,1,\ldots,N\}$$

Discretization of PDE: *PDE evaluated at* (S_n^i, t_n)

$$\frac{\partial v}{\partial t}|_{S_n^i,t_n} + \frac{1}{2}\sigma^2 S^2\frac{\partial^2 v}{\partial S^2}|_{S_n^i,t_n} + rS\frac{\partial v}{\partial S}|_{S_n^i,t_n} - rv|_{S_n^i,t_n} = 0$$

$$\frac{v_{n+1}^i - v_n^i}{\Delta t} + \frac{\sigma^2}{2}(S_n^i)^2\frac{v_n^{i-1} - 2v_n^i + v_n^{i+1}}{h^2} + rS_n^i\frac{v_n^{i+1} - v_n^{i-1}}{2h} - rv_n^i + O(\Delta t + h^2) = 0$$

$$-\frac{1}{2}(\sigma^2 i^2 - ri)\Delta t V_n^{i-1} + (1 + \sigma^2 i^2\Delta t + r\Delta t)V_n^i - \frac{1}{2}(\sigma^2 i^2 + ri)\Delta t V_n^{i+1} = V_{n+1}^i$$

Discretization of terminal and boundary conditions: Dirichlet (put)

$$V_n^0 = (X - S_n^0)e^{-r(T-n\Delta t)} = Xe^{-r(T-n\Delta t)}, V_n^I = 0$$

In general: $a_i V_n^{i-1} + b_i V_n^i + c_i V_n^{i+1} = V_{n+1}^i,\ i \in Z_I = 1,2,\dots,I-1$

$$\begin{bmatrix} a_1 \\ \vdots \\ a_{I-1}\end{bmatrix} \circ \begin{bmatrix} V_n^0 \\ \vdots \\ V_n^{I-2}\end{bmatrix} + \begin{bmatrix} b_1 \\ \vdots \\ b_{I-1}\end{bmatrix} \circ \begin{bmatrix} V_n^1 \\ \vdots \\ V_n^{I-1}\end{bmatrix} + \begin{bmatrix} c_1 \\ \vdots \\ c_{I-1}\end{bmatrix} \circ \begin{bmatrix} V_n^2 \\ \vdots \\ V_n^{I}\end{bmatrix} = \begin{bmatrix} V_{n+1}^1 \\ \vdots \\ V_{n+1}^{I-1}\end{bmatrix}$$

$$\boldsymbol{A}\boldsymbol{V}_n = \boldsymbol{V}_{n+1} + \boldsymbol{F}_n$$

$$\begin{bmatrix}\begin{pmatrix} b_1 & c_1 & & & \\ a_2 & b_2 & c_2 & & \\ & \ddots & \ddots & \ddots & \\ & & a_{I-2} & b_{I-2} & c_{I-2} \\ & & & a_{I-1} & b_{I-1}\end{pmatrix}\end{bmatrix}\begin{bmatrix} V_n^1 \\ V_n^2 \\ \vdots \\ V_n^{I-2} \\ V_n^{I-1}\end{bmatrix} = \begin{bmatrix} V_{n+1}^1 \\ V_{n+1}^2 \\ \vdots \\ V_{n+1}^{I-2} \\ V_{n+1}^{I-1}\end{bmatrix} + \begin{bmatrix} -a_1 V_n^0 \\ 0 \\ \vdots \\ 0 \\ -c_{I-1}V_n^I\end{bmatrix}$$

Obtain $\boldsymbol{V}_n = \boldsymbol{A}^{-1}(\boldsymbol{V}_{n+1} + \boldsymbol{F}_n) = \boldsymbol{A}^{-1}\boldsymbol{V}_{n+1} + \boldsymbol{A}^{-1}\boldsymbol{F}_n$

Monotonicity:

$$A^{-1}U_{n+1} + A^{-1}F_n \le A^{-1}V_{n+1} + A^{-1}F_n, \qquad A^{-1}U_{n+1} \le A^{-1}V_{n+1}$$

Sufficient condition: $\boldsymbol{A} = (a_{ij}),\ a_{ij} \le 0 \ if\ i \ne j,\ a_{ii} > \sum_{i\ne j}|a_{ij}|$

Modified fully implicit scheme:

Discretization of PDE: *PDE evaluated at* (S_n^i, t_n)

$$\frac{\partial v}{\partial t}|_{S_n^i,t_n} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 v}{\partial S^2}|_{S_n^i,t_n} + rS\frac{\partial v}{\partial S}|_{S_n^i,t_n} - rv|_{S_n^i,t_n} = 0$$

$$\frac{v_{n+1}^i - v_n^i}{\Delta t} + \frac{\sigma^2}{2}(S_n^i)^2 \frac{v_n^{i-1} - 2v_n^i + v_n^{i+1}}{h^2} + rS_n^i \frac{v_n^{i+1} - v_n^i}{h} - rv_n^i + \mathcal{O}(\Delta t + h) = 0$$

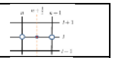
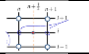
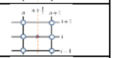
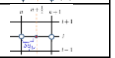
$$-\frac{1}{2}\sigma^2 i^2 \Delta t V_n^{i-1} + \left(1 + \sigma^2 i^2 \Delta t + r\Delta t(1+i)\right)V_n^i - \frac{1}{2}(\sigma^2 i^2 + ri)\Delta t V_n^{i+1} = V_{n+1}^i, \qquad i = 1,2,\dots,I-1$$

Monotone scheme: guaranteed convergence

Crank-Nicolson Scheme

- Evaluate the PDE at "virtual" grid point $t_{n+\frac{1}{2}}$
- Discretize evaluated PDE: centred diff for all derivative

Discretization of PDE: *PDE evaluated at* $\left(S_{n+\frac{1}{2}}^i, t_{n+\frac{1}{2}}\right)$

$v(S,t)\|_{S^i_{n+\frac{1}{2}},t_{n+\frac{1}{2}}} = v^i_{n+\frac{1}{2}} = \frac{1}{2}(v_n^i + v_{n+1}^i) + \mathcal{O}((\Delta t)^2)$	
$\frac{\partial v}{\partial S}\|_{S^i_{n+\frac{1}{2}},t_{n+\frac{1}{2}}} = \frac{1}{2}\left(\frac{v_n^{i+1} - v_n^{i-1}}{2h} + \frac{v_{n+1}^{i+1} - v_{n+1}^{i-1}}{2h}\right) + \mathcal{O}(h^2 + (\Delta t)^2)$	
$\frac{\partial^2 v}{\partial S^2}\|_{S^i_{n+\frac{1}{2}},t_{n+\frac{1}{2}}} = \frac{1}{2}\left(\frac{v_n^{i+1} - 2v_n^i + v_n^{i-1}}{h^2} + \frac{v_{n+1}^{i+1} - 2v_{n+1}^i + v_{n+1}^{i-1}}{h^2}\right) + \mathcal{O}(h^2 + (\Delta t)^2)$	
$\frac{\partial v}{\partial t}\|_{S^i_{n+\frac{1}{2}},t_{n+\frac{1}{2}}} = \frac{v_{n+1}^i - v_n^i}{2\left(\frac{\Delta t}{2}\right)} + \mathcal{O}((\Delta t)^2)$	

$$\frac{v_{n+1}^i - v_n^i}{\Delta t} + \frac{1}{4}\sigma^2\left(S^i_{n+\frac{1}{2}}\right)^2\left(\frac{v_n^{i+1} - 2v_n^i + v_n^{i-1}}{h^2} + \frac{v_{n+1}^{i+1} - 2v_{n+1}^i + v_{n+1}^{i-1}}{h^2}\right)$$

$$+rS^i_{n+\frac{1}{2}}\frac{1}{2}\left(\frac{v_n^{i+1} - v_n^{i-1}}{2h} + \frac{v_{n+1}^{i+1} - v_{n+1}^{i-1}}{2h}\right) - r\left(\frac{v_n^i + v_{n+1}^i}{2}\right) + \mathcal{O}((\Delta t)^2 + h^2) = 0$$

$$\therefore a_i V_n^{i-1} + b_i V_n^i + c_i V_n^{i+1} = \alpha_i V_{n+1}^{i-1} + \beta_i V_{n+1}^i + \gamma_i V_{n+1}^{i+1}, \qquad i = 1,2,\dots,I-1$$

$$\boldsymbol{A}\boldsymbol{V}_n = \boldsymbol{B}\boldsymbol{V}_{n+1} + \boldsymbol{F}_n$$

$$\begin{bmatrix} b_1 & c_1 & & & \\ a_2 & b_2 & c_2 & & \\ & \ddots & \ddots & & \\ & & a_{I-2} & b_{I-2} & c_{I-2} \\ & & & a_{I-1} & b_{I-1}\end{bmatrix}\begin{bmatrix} V_n^1 \\ V_n^2 \\ \vdots \\ V_n^{I-2} \\ V_n^{I-1}\end{bmatrix} = \begin{bmatrix} \beta_1 & \gamma_1 & & & \\ \alpha_2 & \beta_2 & \gamma_2 & & \\ & \ddots & \ddots & & \\ & & \alpha_{I-2} & \beta_{I-2} & \gamma_{I-2} \\ & & & \alpha_{I-1} & \gamma_{I-1}\end{bmatrix}\begin{bmatrix} V_{n+1}^1 \\ V_{n+1}^2 \\ \vdots \\ V_{n+1}^{I-2} \\ V_{n+1}^{I-1}\end{bmatrix} + \begin{bmatrix} \alpha_1 V_{n+1}^0 - a_1 V_n^0 \\ 0 \\ \vdots \\ 0 \\ \gamma_{I-1}V_{n+1}^I - c_{I-1}V_n^I\end{bmatrix}$$

Monotonicity: $\boldsymbol{V}_n = \boldsymbol{A}^{-1}(\boldsymbol{B}\boldsymbol{V}_{n+1} + \boldsymbol{F}_n) \quad \to \quad A^{-1}BU_{n+1} \le A^{-1}BV_{n+1}$

- Not monotone unless $\frac{\Delta t}{h^2}$ is small enough
- Monotonicity sufficient but not necessary for convergence

CN scheme not as stable/robust due to lack of monotonicity compared to modified fully implicit scheme when e.g. non-smooth terminal conditions

Upwind Treatment

$$\frac{\partial u}{\partial \tau} + \varepsilon\frac{\partial^2 u}{\partial x^2} + b\frac{\partial u}{\partial x} = f$$

$$f: known\ term, b\frac{\partial u}{\partial x}: advection\ term, \varepsilon\frac{\partial^2 u}{\partial x^2}: diffusion\ term$$

b: velocity of flow of u. ε: directionless

Applicable when $|b| \gg |\varepsilon|$, apply:

$$\frac{\partial u}{\partial x}|_{x_n^i,t_n} = \begin{cases}\frac{u_n^i = u_n^{i-1}}{\Delta x} + \mathcal{O}(\Delta x), & if\ b > 0 \\ \frac{u_n^{i+1} - u_n^i}{\Delta x} + \mathcal{O}(\Delta x), & if\ b < 0\end{cases}$$

Ensures coefficient matrix in implicit schemes is diagonally dominant

Iterative Methods

$$Ax = b \ \to\ x = D^{-1}(Lx + Ux + b)$$

$$d_{ij} = \begin{cases} a_{ij}, & if\ i = j \\ 0, & otherwise\end{cases}, \quad \ell_{ij} = \begin{cases} -a_{ij}, & if\ i > j \\ 0, & otherwise\end{cases}, \quad u_{ij} = \begin{cases} -a_{ij}, & if\ i < j \\ 0, & otherwise\end{cases}$$

Jacobi iterative method: $x^{k+1} = D^{-1}(Lx^k + Ux^k + b),\ k = 0,1,\dots$

Gauss-Seidel: $x^{k+1} = D^{-1}(Lx^{k+1} + Ux^k + b),\ k = 0,1,\dots$

SOR: $x_{gs}^{k+1} = D^{-1}(Lx^{k+1} + Ux^k + b), x^{k+1} = (1-\omega)x^k + wx_{gs}^{k+1}$

$0 < \omega < 1$: under relax, $\omega = 1$: Gauss-Seidel, $1 < \omega < 2$: over relax

Input $A, x^0, b, M, \varepsilon$ *Output Solution vector* x *Step1 Set* $k = 0$, *Converged* = *False* *Step2 While not Converged* *for* $i = 1,2,\dots,M$ % *compute* $k+1$ $x_i^{k+1} = \frac{1}{a_{ii}}\left[-\sum_{j=1}^{i-1} a_{ij}x_j^k - \sum_{j=i+1}^{M} a_{ij}x_j^k + b_i\right]$ *if* $\Vert x^{k+1} - x^k\Vert < \varepsilon$, *Converged* = *True* $k = k+1$ *Step3 Set* $x = x^k$, *Output*(x)	GS: $x_i^{k+1} = \frac{1}{a_{ii}}\left[-\sum_{j=1}^{i-1} a_{ij}x_j^{k+1} - \sum_{j=i+1}^{M} a_{ij}x_j^k + b_i\right]$ SOR: additional input ω $x_{i,gs}^{k+1} = \frac{1}{a_{ii}}\left[-\sum_{j=1}^{i-1} a_{ij}x_j^{k+1} - \sum_{j=i+1}^{M} a_{ij}x_j^k + b_i\right]$ $x_i^{k+1} = (1-\omega)x_i^k + \omega_{i,gs}^{k+1}$

Non-Smooth Terminal Condition

CS scheme better accuracy than fully implicit, but solution exhibit oscillatory behaviour when terminal condition is non-smooth

Rannacher smoothing: CN scheme with Δt

- Apply CS scheme in time period $[0, T - \Delta t]$
- In $[T - \Delta t, T]$, apply modified fully implicit scheme M times using $\Delta t' = \frac{\Delta t}{M}$. $\mathcal{O}(\Delta t') = \left(\frac{\Delta t}{M}\right)$ in the t dimension

Boundary Conditions

Artificial boundary conditions: achieve bounded domain

Truncation of unbounded domain: $[0, r_{max}]$ for r, at $r = r_{max}$

$b(r_{max}, t) = 0$ for r_{max} sufficiently large, as bond price $\to 0$ as $r \to \infty$

Change of variable: $y = \frac{r}{a-r}, u(y,t) = b(r,t),\ where\ a\ is\ constant$

Solution domain in y is now finite: $y \in [0,1]$ (tut 8)

Chapter 4: Applications of Lattice Methods

Lattice Methods for Lookback Options

$$\frac{\partial v}{\partial t} + \frac{1}{2}\sigma^2 S^2\frac{\partial^2 v}{\partial S^2} + (r-q)S\frac{\partial v}{\partial S} - rv = 0$$

$$0 < S < A, t \in [0,T), \qquad v(S,A,T) = A - S, \qquad \frac{\partial v}{\partial A}(A,A,t) = 0$$

Apply transformation $x = \ln\frac{A}{S},\ w(x,t) = \frac{v(S,A,t)}{S}$

$$\frac{\partial w}{\partial t} + \frac{1}{2}\sigma^2\frac{\partial^2 w}{\partial x^2} + \left(q - r - \frac{\sigma^2}{2}\right)\frac{\partial w}{\partial x} - qw = 0, \qquad x > 0, t \in [0,T)$$

$$w(x,T) = e^x - 1, \qquad \frac{\partial w}{\partial x}(0,t) = 0$$

Explicit difference scheme *for lookback options*

Finite difference grid:

$$\{(x_n^k, t_n\}: x_n^k = k\Delta x, t_n = n\Delta t, n = 0,1,\dots,N, k = -1,0,1,\dots,k_{max}\}$$

Evaluate the PDE at (x_{n+1}^k, t_{n+1}):

$$\frac{w_{n+1}^k - w_n^k}{\Delta t} + \frac{\sigma^2}{2}\frac{w_{n+1}^{k+1} - 2w_{n+1}^k + w_{n+1}^{k-1}}{(\Delta x)^2} + \left(q - r - \frac{\sigma^2}{2}\right)\frac{w_{n+1}^{k+1} - w_{n+1}^{k-1}}{2\Delta x} - qw_n^k + \mathcal{O}(\Delta t + (\Delta x)^2) = 0,$$

$$for\ k = 0,1,\dots$$

$$W_n^k = \frac{1}{1+q\Delta t}[\alpha a W_{n+1}^{k-1} + (1-\alpha)W_{n+1}^k + \alpha(1-a)W_{n+1}^{k+1}],$$

$$k = 0,1,\dots, \qquad \alpha = \sigma^2\frac{\Delta t}{(\Delta x)^2}, \qquad a = \frac{1}{2} - \left(q - r - \frac{\sigma^2}{2}\right)\frac{\Delta x}{2\sigma^2}$$

Single state variable BTM *for lookback options*

$$W_n^0 = e^{-r\Delta t}[puW_{n+1}^0 + (1-p)dW_{n+1}^1],$$

$$W_n^k = e^{-r\Delta t}[puW_{n+1}^{k-1} + (1-p)dW_{n+1}^{k+1}], \qquad k = 1,\dots,n$$

$$W_N^k = u^k - 1 = e^{k\sigma\sqrt{\Delta t}} - 1$$

Set $\alpha = 1 \to \Delta x = \sigma\sqrt{\Delta t}$: $W_n^k = \frac{1}{1+q\Delta t}[aW_{n+1}^{k-1} + (1-a)W_{n+1}^{k+1}],\ k = 0,1,\dots$

$$\therefore e^{-r\Delta t}pu = \frac{a}{1+q\Delta t} + \mathcal{O}(\Delta t), \qquad e^{-r\Delta t}(1-p)d = \frac{1-\alpha}{1+q\Delta t} + \mathcal{O}(\Delta t)$$

- *Link to single-state variable BTM for* $k \ge 1$

Neumann boundary: $0 = \frac{\partial w}{\partial x}|_{0,t_n} = \frac{w_n^0 - w_n^{-1}}{\Delta x} + \mathcal{O}(\Delta x),\ n = 0,1,\dots,N$

$$W_n^{-1} = W_n^0 \to W_n^0 = \frac{1}{1+q\Delta t}[aW_{n+1}^0 + (1-a)W_{n+1}^1]$$

- *Equivalence of FDEs to single-state variable BTM*

$$W_n^0 = \frac{1}{1+q\Delta t}[aW_{n+1}^0 + (1-a)W_{n+1}^1]$$

$$W_n^k = \frac{1}{1+q\Delta t}[aW_{n+1}^{k-1} + (1-a)W_{n+1}^{k+1}], \qquad k = 0,1,\dots$$

$$W_N^k = e^{k\Delta x} - 1, \qquad k = 0,1,\dots$$

How do grids coincide?

Lattice Methods for American Options

LCC at (S_k^i, t_k) *in discretized domain:* $\begin{cases} -\mathcal{L}v|_{S_k^i,t_k} = 0,\ v_k^i - \varphi_k^i > 0 \\ -\mathcal{L}v|_{S_k^i,t_k} \ge 0,\ v_k^i - \varphi_k^i = 0\end{cases}$

Explicit scheme: *discretize* $-\mathcal{L}v|_{S_{n+1}^i,t_{n+1}} \ge 0$

$$\begin{cases} V_n^i - (a_i V_{n+1}^{i-1} + b_i V_{n+1}^i + c_i V_{n+1}^{i-1}) = 0, & V_n^i - \varphi_n^i > 0 \\ V_n^i - (a_i V_{n+1}^{i-1} + b_i V_{n+1}^i + c_i V_{n+1}^{i-1}) \ge 0, & V_n^i - \varphi_n^i = 0\end{cases}$$

$$min\{V_n^i - (a_i V_{n+1}^{i-1} + b_i V_{n+1}^i + c_i V_{n+1}^{i-1}), V_n^i - \varphi_n^i\} = 0$$

$$V_n^i = max\{a_i V_{n+1}^{i-1} + b_i V_{n+1}^i + c_i V_{n+1}^{i-1}, \varphi_n^i\}$$

$$n = N-1,\dots,0, \qquad i = 1,\dots,I-1, \qquad V_N^i = \varphi(S_N^i), i = 0,1,\dots,I$$

Implicit scheme: *discretize* $-\mathcal{L}v|_{S_n^i,t_n} \ge 0$

$$\begin{cases}(a_i V_n^{i-1} + b_i V_n^i + c_i V_n^{i-1}) - (\alpha_i V_{n+1}^{i-1} + \beta_i V_{n+1}^i + \gamma_i V_{n+1}^{i-1}) = 0, & V_n^i - \varphi_n^i > 0 \\ (a_i V_n^{i-1} + b_i V_n^i + c_i V_n^{i-1}) - (\alpha_i V_{n+1}^{i-1} + \beta_i V_{n+1}^i + \gamma_i V_{n+1}^{i-1}) \ge 0, & V_n^i - \varphi_n^i > 0\end{cases}$$

$$\begin{cases} V_n^i - \frac{1}{b_i}[(-a_i V_n^{i-1} - c_i V_n^{i+1}) + (\alpha_i V_{n+1}^{i-1} + \beta_i V_{n+1}^i + \gamma_i V_{n+1}^{i+1})] = 0, & V_n^i - \varphi_n^i > 0 \\ V_n^i - \frac{1}{b_i}[(-a_i V_n^{i-1} - c_i V_n^{i+1}) + (\alpha_i V_{n+1}^{i-1} + \beta_i V_{n+1}^i + \gamma_i V_{n+1}^{i+1})] \ge 0, & V_n^i - \varphi_n^i > 0\end{cases}$$

$$min\left\{V_n^i - \frac{1}{b_i}[(-a_i V_n^{i-1} - c_i V_n^{i+1}) + (\alpha_i V_{n+1}^{i-1} + \beta_i V_{n+1}^i + \gamma_i V_{n+1}^{i+1})], V_n^i - \varphi_n^i\right\} = 0$$

$$V_n^i = max\left\{\frac{1}{b_i}[(-a_i V_n^{i-1} - c_i V_n^{i+1}) + (\alpha_i V_{n+1}^{i-1} + \beta_i V_{n+1}^i + \gamma_i V_{n+1}^{i+1})], \varphi_n^i\right\}$$

Each iteration, a value first obtained from scaler form implicit FDE, compared with immediate payoff. Max of 2 taken to yield new iterate

PSOR *Input* $A, b, V_{n+1}, M, \varepsilon, \omega, \varphi_n$ *Output Solution vector* V_n *Step1 Set* $k = 0$, *Converged* = *False*, $x^0 = V_{n+1}$ *Step2 While not Converged*   *for* $i = 1,2,\dots,M$ % *compute iterate* $k+1$ $x_{i,gs}^{k+1} = \frac{1}{a_{ii}}\left[-\sum_{j=1}^{i-1} a_{ij}x_j^{k+1} - \sum_{j=i+1}^{M} a_{ij}x_j^k + b_i\right]$ $x_i^{k+1} = max\left((1-\omega)x_i^k + \omega x_{i,gs}^{k+1}, \varphi_n^i\right)$ *If* $\Vert x^{k+1} - x^k\Vert < \varepsilon$ *Then set Converged* = *True*   *Set* $k = k+1$ *Step3 Set* $V_n = x^k$

Option with 2 Underlying Assets

$$\frac{\partial v}{\partial t} + \frac{1}{2}\sigma_1^2 S_1^2\frac{\partial^2 v}{\partial S_1^2} + \rho\sigma_1\sigma_2 S_1 S_2\frac{\partial^2 v}{\partial S_1\partial S_2} + \frac{1}{2}\sigma_2^2 S_2^2\frac{\partial^2 v}{\partial S_2^2} + (r-q_1)S_1\frac{\partial v}{\partial S_1} + (r-q_2)S_2\frac{\partial v}{\partial S_2} - rv = 0$$

In domain $\{S_1 > 0, S_2 > 0, t \in [0,T)\}$ *and subject to terminal condition* $v(S_1,S_2,T) = \varphi(S_1,S_2)$

Discretization of domain:

Truncate domain $\{0 \le S_1 \le S_1^{max}, 0 \le S_2 \le S_2^{max}, 0 \le t \le T\}$

Grid: Choose sufficiently large integers I and J and define grid spacings $h_1 = S_1^{max}/I$, $h_2 = S_2^{max}/J$

Then grid is given by $\{(S_{1,n}^i, S_{2,n}^j, t_n): S_{1,n}^i = ih_1, S_{2,n}^j = jh_2, t_n = n\Delta t\ i = 0,\dots,I, j = 0,\dots,J, n = 0,\dots,N\}$

Denote $v_n^{i,j} = v(S_{1,n}^i, S_{2,n}^j, t_n)$ and $V_n^{i,j} = V(S_{1,n}^i, S_{2,n}^j, t_n)$

Discretizing Cross Derivative

$$\frac{\partial^2 v}{\partial S_1\partial S_2}(S_{1,n}^i, S_{2,n}^j, t_n) = \frac{v_n^{i+1,j+1} + v_n^{i-1,j-1} - v_n^{i+1,j-1} - v_n^{i-1,j+1}}{\boldsymbol{4h_1h_2}} + O(h_1^2 + h_2^2)$$

Boundary Conditions: Let $P_E(S,\sigma,q,t)$ be price function of European vanilla put.

$$v(0,S_2,t) = Xe^{-r(T-t)}, v(S_1,0,t) = Xe^{-r(T-t)}, v(S_1^{max},S_2,t) = P_E(S_2,\sigma_2,q_2,t), v(S_1,S_2^{max},t) = P_E(S_1,\sigma_1,q_1,t)$$

FDE Formula: Applying Forward Time formula

$$\frac{v_{n+1}^{i,j} - v_n^{i,j}}{\Delta t} + \frac{\sigma_1^2}{2}i^2h_1^2\frac{v_n^{i-1,j} - 2v_n^{i,j} + v_n^{i+1,j}}{h_1^2} + \rho\sigma_1\sigma_2 ijh_1h_2\frac{v_n^{i+1,j+1} + v_n^{i-1,j-1} - v_n^{i+1,j-1} - v_n^{i-1,j+1}}{4h_1h_2}$$

$$+\frac{\sigma_2^2}{2}j^2h_2^2\frac{v_n^{i,j-1} - 2v_n^{i,j} + v_n^{i,j+1}}{h_2^2} + (r-q_1)ih_1\frac{v_n^{i+1,j} - v_n^{i-1,j}}{2h_1}$$

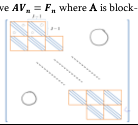
$$+(r-q_2)jh_2\frac{v_n^{i,j+1} - v_n^{i,j-1}}{2h_2} - rv_n^{i,j} + O(\Delta t + h_1^2 + h_2^2) = 0$$

Which simplifies to ($\alpha_i = \frac{\sigma_1^2 i^2\Delta t}{2}, \beta_i = (r-q_1)i\Delta t/2, \gamma = \rho\sigma_1\sigma_2 ij\Delta t/4$) (for j replace: $i = j, \sigma_1 = \sigma_2, q_1 = q_2$)

$$-\gamma V_n^{i-1,j-1} - (\alpha_j - \beta_j)V_n^{i,j-1} + \gamma V_n^{i+1,j-1} - (\alpha_i - \beta_i)V_n^{i-1,j} + (1 + 2\alpha_i + 2\alpha_j - r\Delta t)V_n^{i,j} - (\alpha_i + \beta - i)V_n^{i+1,j}$$

$$+\gamma V_n^{i-1,j+1} - (\alpha_j + \beta_j)V_n^{i,j+1} - \gamma V_n^{i+1,j+1} = V_{n+1}^{i,j}$$

Denote $\boldsymbol{V}_n = [V_n^{1,1}, V_n^{1,2}, \dots, V_n^{I-1,J-1}]^T$ we have $\boldsymbol{AV_n} = \boldsymbol{F_n}$ where **A** is block-tridiagonal with $(J-1)(J-1)$ blk



Chapter 5: Monte Carlo Methods

Euler Approximation

No Transformation Single Asset:

$$dS_t = (r-q)S_t dt + \sigma S_t dB_t \to \Delta S_n = (r-q)S_n\Delta t + \sigma S_n\varepsilon_n\sqrt{\Delta t}$$

For $\{n = 0,\dots,N-1\}$ where $\Delta S_n = S_{n+1} - S_n$ and $\varepsilon_n \sim N(0,1)$

$$S_{n+1} = S_n + (r-q)S_n\Delta t + \sigma S_n\varepsilon_n\sqrt{\Delta t} = S_n\left(1 + (r-q)\Delta t + \sigma\varepsilon_n\sqrt{\Delta t}\right)$$

Transformation: $x = \ln S$

$$dx = \left(r - q - \frac{\sigma^2}{2}\right)dt + \sigma dB \to \ln S_{n+1} - \ln S_n = \left(r - q - \frac{\sigma^2}{2}\right)\Delta t + \sigma\varepsilon_n\sqrt{\Delta t}$$

$$S_{n+1} = S_n\exp\left(\left(r - q - \frac{\sigma^2}{2}\right)\Delta t + \sigma\varepsilon_n\sqrt{\Delta t}\right)$$

Generating Asset Prices

Simple Method $(\psi \sim Uniform(0,1))$	**Box-Muller method** $(x_1, x_2 \sim iid\ Uniform(0,1))$
$z \approx -6 + \Sigma_{i=1}^{12}\psi_i$	$y_1 = \sqrt{-2\ln(x_1)}\cos(2\pi x_2)$ $y_2 = \sqrt{-2\ln(x_1)}\sin(2\pi x_2)$

Standard Error of Estimates and Number of Paths (Square root of P rule)

Let f be discounted payoff of security, $\bar{f}$ the true estimate ($\bar{f} = E(f), \sigma_f^2 = var(f)$)

$$\bar{f}_P = \frac{1}{P}\Sigma_{i=1}^P f_i, s_P^2 = \frac{1}{P-1}\Sigma_{i=1}^P(f_i - \bar{f}_P)^2$$

Error $\bar{f}_P - \bar{f} \sim N(0, \sigma_e^2)$ where the standard error $\sigma_e = \sigma_f/\sqrt{P}$ (error is inversely proportional to $\sqrt{P}$)

Simulations for Path-Dependent Options (European-Style)

Barrier	Maintain indicator value for each simulated path to flag status at expiry
Asian	Maintain average values along simulated path to evaluate at expiry
Lookback	Maintain max/min of prices along each simulated path

Generating Correlated Sample Prices for Multi-Asset Options (d correlated assets)

1. Use Cholesky decomposition on correlation matrix $\boldsymbol{C} = \boldsymbol{LL}^t$ ($\boldsymbol{L} = \begin{bmatrix} 1 & 0 \\ \rho_{12} & \sqrt{1-\rho_{12}^2}\end{bmatrix}$ for 2 asset)
2. Draw d independent samples from standard normal and form vector $\boldsymbol{x} \coloneqq (x_i)$ $(\sim N(0,I))$
3. Correlated samples $\varepsilon_i, i = 1,\dots,d$ given by $\boldsymbol{\varepsilon} = \boldsymbol{Lx}$
4. Simulated Price for Asset i: $S_{i,n+1} = S_{i,n}\exp\left(\left(r - q_i - \frac{\sigma_i^2}{2}\right)\Delta t + \sigma_i\varepsilon_{i,n}\sqrt{\Delta t}\right)$

$\varepsilon_{1,n} = x_{1,n}, \varepsilon_{2,n} = \rho_{12}x_{1,n} + \sqrt{1-\rho_{12}^2}x_{2,n}, \varepsilon_{3,n} = \rho_{13}x_{1,n} + \frac{\rho_{23} - \rho_{13}\rho_{12}}{\sqrt{1-\rho_{12}^2}}x_{2,n} + \sqrt{\frac{1 + 2\rho_{23}\rho_{12}\rho_{13} - \rho_{12}^2 - \rho_{13}^2 - \rho_{23}^2}{1-\rho+12\ \hat{}2}}x_{3,n}$ (d=3)

Variance Reduction Techniques

Antithetic Variance Reduction: Generate $P/2$ paths, for each path, also consider mirrored path. (+) No extra computation needed but (-) has limited effect on reducing standard error.

Control Variate: Suppose there is derivative B that has (1) price F_B correlated with F_A, and (2) $E[F_b]$ is available in closed form, then can use F_B as control variate. Choose $B^* = \frac{\sigma(F_A)}{\sigma(F_B)}\rho_{A,B} = \frac{cov(F_A,F_B)}{var(F_B)}$ then the variance is given by $var(F_A^{B^*}) = (1 - \rho_{A,B}^2)var(F_A)$. In practice, B^* is obtained using same run of MC simulations to obtain $\overline{F_A}$ and $\overline{F_B}$

$$\hat{B}^* = \frac{\Sigma_{i=1}^P(F_{A,i} - \overline{F_A})(F_{B,i} - \overline{F_B})}{\Sigma_{i=1}^P(F_{B,i} - \overline{F_B})^2} = \frac{sample\ covariance}{sample\ variance}$$

Example of Control Variate: Consider European maximum call $\varphi(S_1,S_2) = (\max(S_1,S_2), X)^+$ then at expiry, payoff is either $(S_1 - X)^+$ or $(S_2 - X)^+$ so portfolio with equal weighted vanilla calls can be expected to have good correlation with two-asset option and can be used as control variate with payoff $\frac{1}{2}(S_1 - X)^+ + \frac{1}{2}(S_2 - X)^+$, which has closed-form solution to control variate given by $\frac{1}{2}C_E(S_1,\sigma_1,q_1,t) + \frac{1}{2}C_E(S_2,\sigma_2,q_2,t)$

Chapter 6: Beyond Black-Scholes

Flaws in Black-Scholes model

Implied Volatility Volatility represents uncertainty in underlier's return and can be estimated from historical prices. Using BSM, we can instead use the market price as input to "reverse-engineer" the implied volatility

$$g(\sigma) \coloneqq e^{-qT}S_0N(d_1) - Xe^{-rT}N(d_2) - C_{market} = 0$$

Using a numerical method to solve for root (e.g. bisection search)

Flaws (1) BSM assumes underlying asset price path is continuous but real-life scenarios show price can "jump" from time to time. (2) BSM assumes delta-hedging is continuous, but there's time interval between re-hedges, so risk is not fully eliminated. (3) BSM assumes no costs incurred in delta hedging, but one must consider transaction cost incurred since high costs makes frequent hedging costly.

Local Volatility Model

Treat volatility to be a deterministic function of S and t, then we have

$$\frac{dS}{S} = \mu dt + \sigma(S,t)dW, \frac{\partial v}{\partial t} + \frac{1}{2}\sigma^2(S,t)S^2\frac{\partial^2 v}{\partial S^2} + (r-q)S\frac{\partial v}{\partial S} - rv = 0$$

In general, no closed form solution, so require numerical methods such as FDS (discretize σ) or MC.

Special Case: $\sigma = \sigma(t)$ (volatility is function of time only), we replace σ in BSM with

$$\sqrt{\frac{1}{T-t}\int_t^T \sigma^2(\tau)d\tau} \to v(S_t,t) = \begin{cases} e^{-q(T-t)}S_tN(d_1) - Xe^{-r(T-t)}N(d_2), (call) \\ Xe^{-r(T-t)}N(-d_2) - e^{-q(T-t)}S_tN(-d_1), (put)\end{cases}$$

$$d_1 = \frac{\ln\left(\frac{S_t}{X}\right) + (r-q)(T-t) + \frac{1}{2}\int_t^T\sigma^2(\tau)d\tau}{\sqrt{\int_t^T\sigma^2(\tau)d\tau}}, d_2 = d_1 - \sqrt{\int_t^T\sigma^2(\tau)d\tau}$$

Extension We can also assume $r = r(t)$ and $q = q(t)$, replacing $r(T-t)$ and $q(T-t)$ with

$$\int_t^T r(\tau)d\tau\ and\ \int_t^T q(\tau)d\tau$$

Calibration $\sigma(S,t)$ can be obtained via calibration. For example, different option values and different term to maturity options can be used to obtain $\sigma(t)$ by curve-fitting

Stochastic Volatility Model

Treat volatility to be a random variable:

$$d\sigma_t = p(S_t,\sigma_t,t)dt + g(S_t,\sigma_t,t)dW_{2t}, dS_t = \mu S_t dt + \sigma_t S_t dW_{1t}$$

Where two Wiener increments have correlation of ρ ($dW_{1t}dW_{2t} = \rho dt\ in\ SDE$), option value is $v(S,\sigma,t)$

Delta-Hedging Source of risk σ is not traded asset so we use another option with same underlier as proxy for σ, with value $\bar{v}(S,\sigma,t)$.

Hedging portfolio value: $\pi = v - \Delta_1 S - \Delta_2\bar{v}$, $d\pi = dv - \Delta_1 dS - \Delta_2 d\bar{v}$, then apply Ito's lemma twice:

$$d\pi = \left(\frac{\partial v}{\partial t} + \frac{1}{2}\sigma^2S^2\frac{\partial^2 v}{dS^2} + \rho\sigma gS\frac{\partial^2 v}{\partial S\partial\sigma} + \frac{1}{2}g^2\frac{\partial^2 v}{\partial\sigma^2}\right)dt - \Delta_2\left(\frac{\partial\bar{v}}{\partial t} + \frac{1}{2}\sigma^2S^2\frac{\partial^2\bar{v}}{dS^2} + \rho\sigma gS\frac{\partial^2\bar{v}}{\partial S\partial\sigma} + \frac{1}{2}g^2\frac{\partial^2\bar{v}}{\partial\sigma^2}\right)dt$$

$$+\left(\frac{\partial v}{\partial S} - \Delta_2\frac{\partial\bar{v}}{\partial S} - \Delta_1\right)dS + \left(\frac{\partial v}{\partial\sigma} - \Delta_2\frac{\partial\bar{v}}{\partial\sigma}\right)d\sigma$$

Eliminating two source of randomness requires

$$\Delta_1 = \frac{\partial v}{\partial S} - \Delta_2\frac{\partial\bar{v}}{\partial S}, \Delta_2 = \frac{\partial v}{\partial\sigma}\Big/\frac{\partial\bar{v}}{\partial\sigma}$$

By no-arbitrage argument, we have

$$d\pi = \left(\frac{\partial v}{\partial t} + \frac{1}{2}\sigma^2S^2\frac{\partial^2 v}{dS^2} + \rho\sigma gS\frac{\partial^2 v}{\partial S\partial\sigma} + \frac{1}{2}g^2\frac{\partial^2 v}{\partial\sigma^2}\right)dt - \Delta_2\left(\frac{\partial\bar{v}}{\partial t} + \frac{1}{2}\sigma^2S^2\frac{\partial^2\bar{v}}{dS^2} + \rho\sigma gS\frac{\partial^2\bar{v}}{\partial S\partial\sigma} + \frac{1}{2}g^2\frac{\partial^2\bar{v}}{\partial\sigma^2}\right)dt = r\pi dt$$

$$= r(v - \Delta_1 S - \Delta_2\bar{v})dt = r\left(v - S\frac{\partial v}{\partial S} + \Delta_2 S\frac{\partial\bar{v}}{\partial S} - \Delta_2\bar{v}\right)dt$$

Simplifying into

$$\left(\frac{\partial v}{\partial t} + \frac{1}{2}\sigma^2S^2\frac{\partial^2 v}{dS^2} + \rho\sigma gS\frac{\partial^2 v}{\partial S\partial\sigma} + \frac{1}{2}g^2\frac{\partial^2 v}{\partial\sigma^2} + rS\frac{\partial v}{\partial S} - rv\right)dt$$

$$-\Delta_2\left(\frac{\partial\bar{v}}{\partial t} + \frac{1}{2}\sigma^2S^2\frac{\partial^2\bar{v}}{dS^2} + \rho\sigma gS\frac{\partial^2\bar{v}}{\partial S\partial\sigma} + \frac{1}{2}g^2\frac{\partial^2\bar{v}}{\partial\sigma^2} + rS\frac{\partial\bar{v}}{\partial S} - r\bar{v}\right)dt = 0$$

And after collecting the terms, we get

$$\frac{\frac{\partial v}{\partial t} + \frac{1}{2}\sigma^2S^2\frac{\partial^2 v}{dS^2} + \rho\sigma gS\frac{\partial^2 v}{\partial S\partial\sigma} + \frac{1}{2}g^2\frac{\partial^2 v}{\partial\sigma^2} + rS\frac{\partial v}{\partial S} - rv}{\frac{\partial v}{\partial\sigma}} = \frac{\frac{\partial\bar{v}}{\partial t} + \frac{1}{2}\sigma^2S^2\frac{\partial^2\bar{v}}{dS^2} + \rho\sigma gS\frac{\partial^2\bar{v}}{\partial S\partial\sigma} + \frac{1}{2}g^2\frac{\partial^2\bar{v}}{\partial\sigma^2} + rS\frac{\partial\bar{v}}{\partial S} - r\bar{v}}{\frac{\partial\bar{v}}{\partial\sigma}}$$

Which is only possible if both sides of equation are independent of contract type and only function of independent variable S,σ,t (i.e. $-a(S,\sigma,t)$). This yields the final form with problem domain $\{S > 0, \sigma > 0, t \in [0,T]\}$

$$\frac{\partial v}{\partial t} + \frac{1}{2}\sigma^2S^2\frac{\partial^2 v}{dS^2} + \rho\sigma gS\frac{\partial^2 v}{\partial S\partial\sigma} + \frac{1}{2}g^2\frac{\partial^2 v}{\partial\sigma^2} + rS\frac{\partial v}{\partial S} + a(S,\sigma,t)\frac{\partial v}{\partial\sigma} - rv = 0, \qquad v(S,\sigma,T) = \begin{cases}(S-X)^+, call \\ (X-S)^+, put\end{cases}$$

Market Price of Risk

We consider risk-neutral world where

$$dS = rSdt + \sigma SdW_1, d\sigma = a(S,\sigma,t)dt + g(S,\sigma,t)dW_2$$

with $a(S,\sigma,t) = p(S,\sigma,t) - \lambda(S,\sigma,t)g(S,\sigma,t)$ where $\lambda(S,\sigma,t)$ is market price of risk

Heston's Stochastic Volatility Model

Heston propose that volatility $\sigma(t)$ follows the process $d\sigma = -\beta\sigma + \delta dW$ and process of variance can be derived using Ito's Lemma to get $d\sigma^2 = (\delta^2 - 2\beta\sigma^2)dt + 2\delta\sigma dW$ which is in mean-reversion form $d\sigma^2 = k(b - \sigma^2)dt + c\sigma dW$ where k, b, c are constants.

Algorithm 9: Stochastic Volatility Model MC Simulation ($d\sigma_t = (p - \lambda g)dt + gdW_{2,t}$)

Inputs: $\boldsymbol{S_0}, X, r, T, \boldsymbol{\sigma}, \boldsymbol{q}, N, \rho, P, \lambda, p, g$

Compute: $\Delta t = \frac{T}{N}$

Generate $2NP$ standard normal variates x_{1Ns}, x_{2Ns} for $n = 1,\dots,N,\ s = 1,\dots,P$

Transform the standard normal variates into $N \times P$ bivariate normal random variates

$(\varepsilon_{1ns}, \varepsilon_{2ns}) \sim N\left(\boldsymbol{0}, \begin{pmatrix} 1 & \rho \\ \rho & 1\end{pmatrix}\right)$ *for each* (n,s)

for s = 1 to P do:
- $S_{0,s} = S_0$
- $\sigma_{0,s} = \sigma_0$
- for $n = 0$ to $N-1$ do:
  - $S_{n+1,s} = S_n + rS_n\Delta t + \sigma_n S_n\sqrt{\Delta t}\varepsilon_{1ns}$
  - $\sigma_{n+1,s} = \sigma_n + (p - \lambda g)\Delta t + g\sqrt{\Delta t}\varepsilon_{2ns}$
- $V_s = (X - S_{N,s})^+$

Output: $mean(V)$

Jump-Diffusion Model

Poisson Process $N(t)$ is a counting process with rate or intensity $\lambda > 0$ if (1) $N(0) = 0$, (2) $N(t)$ has independent and stationary increments, (3) $P[N(t+dt) - N(t) = 1] = \lambda dt + o(dt)$, (4) $P[N(t+dt) - N(t) \ge 2] = o(dt)$ where $o(dt)$ tends to zero faster than dt

Define jump arrival process dp as

$$dp = \begin{cases} 1, with\ probability\ \lambda dt \\ 0, with\ probability\ 1 - \lambda dt\end{cases}, \quad E(dp) = \lambda dt, \qquad var(dp) = \lambda dt(1 - \lambda dt)$$

Jump-Diffusion Process Geometric Brownian Motion incorporating jumps is given by

$$\frac{dS}{S} = \mu dt + \sigma dW + (J-1)dp$$

Where J is positive random variable controlling size of jump (e.g. $J = 0.9$ represents 10% fall in S)

Using Ito's Lemma, we get

$$dv = \left(\frac{\partial v}{\partial t} + \frac{1}{2}\sigma^2S^2\frac{\partial^2 v}{\partial S^2} + \mu S\frac{\partial v}{\partial S}\right)dt + \sigma S\frac{\partial v}{\partial S}dW + (v(JS,t) - v(S,t))dp$$

Hedging with Jumps Hedging Portfolio has value at time t, $\pi = v(S,t) - \Delta S$

$$d\pi = dv - \Delta dS = \left(\frac{\partial v}{\partial t} + \frac{1}{2}\sigma^2S^2\frac{\partial^2 v}{\partial S^2} + \mu S\frac{\partial v}{\partial S}\right)dt + \sigma S\frac{\partial v}{\partial S}dW + (v(JS,t) - v(S,t))dp$$

$$-\Delta(\mu Sdt + \sigma SdW + (J-1)Sdp)$$

$$= \left(\frac{\partial v}{\partial t} + \frac{1}{2}\sigma^2S^2\frac{\partial^2 v}{\partial S^2} + \mu S\left(\frac{\partial v}{\partial S} - \Delta\right)\right)dt + \sigma S\left(\frac{\partial v}{\partial S} - \Delta\right)dW + (v(JS,t) - v(S,t) - \Delta(J-1)S)dp$$

$$\Delta = \frac{\partial v}{\partial S}, \qquad d\pi = \left(\frac{\partial v}{\partial t} + \frac{1}{2}\sigma^2S^2\frac{\partial^2 v}{\partial S^2}\right)dt + (v(JS,t) - v(S,t) - \Delta(J-1)S)dp$$

Merton's Jump Diffusion Model If jump is uncorrelated with market and hence diversifiable, should not be priced so $E(d\pi) = r\pi dt$. Since $E(dp) = \lambda dt$

$$E((v(JS,t) - v(S,t) - \Delta(J-1)S)dp) = \lambda\left(v(JS,t) - v(S,t) - \Delta(J-1)S\frac{\partial v}{\partial S}\right)dt$$

$$\frac{\partial v}{\partial t} + \frac{1}{2}\sigma^2S^2\frac{\partial^2 v}{\partial S^2} + rS\frac{\partial v}{\partial S} - rv + \lambda(v(JS,t) - v(S,t)) - \lambda(J-1)S\frac{\partial v}{\partial S} = 0\ (J\ is\ deterministic)$$

$$\frac{\partial v}{\partial t} + \frac{1}{2}\sigma^2S^2\frac{\partial^2 v}{\partial S^2} + rS\frac{\partial v}{\partial S} - rv + \lambda E_J(v(JS,t) - v(S,t)) - \lambda E_J(J-1)S\frac{\partial v}{\partial S} = 0\ (J\ is\ random\ quantity)$$

PDE has closed-form solution for (1) European-styled options if (2) J is lognormally distributed ($J \sim logN(\mu_J, \sigma_J^2)$), denoting $\kappa = E_J(J-1)$, if $\ln(J) \sim N(\mu_J, \sigma_J^2)$ then $\kappa = \exp\left(\mu_J + \frac{\sigma_J^2}{2}\right) - 1$

$$\frac{dS}{S} = \mu dt + \sigma dW + (J-1)dp\ (GBM\ incorpating\ jumps)$$

$$S(t) = S(0)\exp\left(\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma W(t)\right)\prod_{i=1}^{N(t)} J_i\ (Solution\ to\ GBM\ incorpating\ jumps)$$

$$S_{n+1} = S_0\exp\left(\left(\mu - \frac{\sigma^2}{2}\right)t_{n+1} + \sigma W(t_{n+1})\right)\prod_{i=1}^{N(t_{n+1})} J_i$$

$$= S_n\exp\left(\left(\mu - \frac{\sigma^2}{2}\right)(t_{n+1} - t_n) + \sigma(W(t_{n+1}) - W(t_n))\right)\prod_{i=N(t_n)+1}^{N(t_{n+1})} J_i$$

For risk neutral pricing, drift parameter $\mu = r - q - \lambda\kappa$

Monte Carlo Simulation of Asset Prices with Jumps

If simulated prices are required at a set of fixed time points, $0 = t_0 < t_1 < \cdots < t_N$ then we can use the following expression to obtain $S_1, S_2, \dots, S_N$

$$S_{n+1} = S_n\exp\left(\left(\mu - \frac{\sigma^2}{2}\right)(t_{n+1} - t_n) + \sigma\varepsilon_n\sqrt{t_{n+1} - t_n}\right)\prod_{i=N(t_n)+1}^{N(t_{n+1})} J_i$$

Alternatively, we can use log price form with $X_n \coloneqq \ln(S_n)$, noting that $\ln(J_i) \sim N(\mu_J, \sigma_J^2)$

$$X_{n+1} = X_n + \left(\mu - \frac{\sigma^2}{2}\right)(t_{n+1} - t_n) + \sigma\varepsilon\sqrt{t_{n+1} - t_n} + \sum_{i=N(t_n)+1}^{N(t_{n+1})}\ln(J_i)$$

Algorithm 10: Jump-Diffusion Monte Carlo for European up-and-out barrier call option

Inputs: $S_0, X, r, T, \sigma, q, N, \lambda, a, b, P, H$

Compute: $\Delta t = \frac{T}{N}, \kappa = \exp\left(a + \frac{b^2}{2}\right) - 1, \mu = r - q - \lambda\kappa$

for $j = 1$ to P do:
- $S(j,0) = S_0$
- for $n = 0$ to $N-1$ do:
  - Generate Random Standard Normal $\epsilon \sim N(0,1)$
  - Generate K from Poisson distribution, $Poisson(\lambda\Delta t)$
  - If $K > 0$ then
    - Generate $\ln(J_1), \dots, \ln(J_K)$ from $N(a, b^2)$ and set $Q \coloneqq \Sigma_{i=1}^K\ln(J_i)$
  - Else
    - $Q \coloneqq 0$
  - $X \coloneqq \ln(S(j,n)) + \left(\mu - \frac{\sigma^2}{2}\right)\Delta t + \sigma\epsilon\sqrt{\Delta t} + Q$
  - $S(j,n+1) \coloneqq \exp(X)$
- $M \coloneqq \max_{0\le n\le N} S(j,n)$
- If $M < H$
  - $V(j) \coloneqq \max(S(j,N) - X, 0)$
- Else
  - $V(j) \coloneqq 0$

Output: $\exp(-rT) \times mean(V)$

Algorithm 11: Multi-Asset MC Simulation

Inputs: $\boldsymbol{S_0}, X, r, T, \boldsymbol{\sigma}, \boldsymbol{q}, N, \rho, P$

Compute: $\Delta t = \frac{T}{N}$

Generate $2P$ standard normal variates x_{1s}, x_{2s} for $n = 1,\dots,N,\ s = 1,\dots,P$

Transform the standard normal variates into P bivariate normal random variates

$(\varepsilon_{1s}, \varepsilon_{2s}) \sim N\left(\boldsymbol{0}, \begin{pmatrix} 1 & \rho \\ \rho & 1\end{pmatrix}\right), s = 1,\dots,P$

for s = 1 to P do:
- $S_{1,T,s} = S_{0,1}\exp\left((r - q - \sigma_1^2/2)T + \sigma_1\sqrt{T}\varepsilon_{1s}\right)$
- $S_{2,T,s} = S_{0,2}\exp\left((r - q - \sigma_2^2/2)T + \sigma_2\sqrt{T}\varepsilon_{2s}\right)$
- $V_s = e^{-rT}\varphi(S_{1,T,s}, S_{2,T,s})$
- for $n = 0$ to $N-1$ do:

Output: $mean(V)$

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